

crew compartment above the ingress-egress hatch and between the windows and two more control panels mounted on the side walls.

At the base of the assembly was the ascent engine – a fixed, constant-thrust rocket with a thrust of about 15,000 N. The stage also contained an aerzine 50 fuel and an oxidizer tank, and helium, liquid oxygen, gaseous oxygen, and reaction control fuel tanks. The dry mass of the ascent stage was 2180 kg and it held 2639 kg of propellant.

One small step...

At 20:17:40 on July 20, 1969, partially piloted manually by Armstrong, the Eagle landed in the Sea of Tranquility at 0.6741 degrees N latitude, 23.4730 degrees E longitude. Armstrong's calm voice came over the radio, "Houston, Tranquility Base here – the Eagle has landed."

After a short rest period, Armstrong stepped onto the lunar surface at 02:56:15 on July 21, stating, "That's one small step for man, one giant leap for mankind." He then collected a small contingency sample of lunar material.

Aldrin followed 19 minutes later, calling the lunar surface "magnificent desolation". The astronauts then unveiled the plaque mounted on a strut behind the ladder and read the inscription aloud: "Here men from the planet Earth first set foot on the Moon, July 1969, A.D. We came in peace for all mankind." They put up an American flag and talked to President Nixon by radiotelephone.

During the EVA, in which they both ranged up to 100 metres from the Eagle, Aldrin deployed the Early Apollo Scientific Experiments Package, or EASEP, experiments.

The EASEP consisted of a set of scientific instruments emplaced at the Apollo 11 landing site by the astronauts. This package was the forerunner of the ALSEP (Apollo Lunar Surface Scientific Experiments Package) used on the later Apollo missions. It consisted of two solar panels to provide power (the EASEP could only operate during lunar day), an antenna and communications system to send data to Earth ground stations and receive commands, a passive seismometer, designed to measure seismic activity and physical properties of the lunar crust and interior, and a lunar dust detector, to measure dust accumulation and radiation damage to solar cells.

The whole contraption comprised a square base on which was mounted the seismometer and dust detector, along with an isotope heater and cylindrical antenna mast with an antenna positioning mechanism. Two brackets protruded from opposite sides of the base and held the canted rectangular